Web SDK & b-rite — Comprehensive Guide

One-pager for AI-assisted web app development. Next.js + Supabase + Vercel.

Tech Stack & AI Agents

Layer	Technology
Framework	Next.js 14 (App Router)
Database & Auth	Supabase (PostgreSQL, RLS, auth)
Hosting	Vercel
Client	React, TypeScript strict, Zod validation
AI Agents	Cursor, Claude (primary). b-rite scripts drive AI workflow.

> **Industry standards:** TypeScript strict, Zod for validation, RLS on all tables, migrations-only schema changes. No `any` types. Security: ai/specs/security.md is authoritative; never relax RLS without explicit instruction.

How the AI Stays on Track

Protected files (br-ignore.br): b-rite.md, brain.md, error handlers, help.br cannot be altered by standard or *dev* calls. Only !ADMIN! *dev* can touch them.

Spec-driven: AI reads ai/specs/ (product, architecture, database, security) before generating code. Migrations go through db/migrations/; schema.sql is never edited directly.

Validation: interMode = 0 enforces strict syntax; AI halts on malformed scripts. guessingMode = 0 forces AI to ask on ambiguity.

b-rite Language — Brief

b-rite is an **AI Nav DSL** (AI Navigation Domain-Specific Language). It does not execute code; it tells the AI where to look, what to read, and what to do. Fully optional — every action can be done with plain-text prompts.

Purpose: Minimize prompt length, standardize workflows, prevent drift. Scripts open with `br :` and close with `:`. Commands: read, fetch, write, ask, prompt, catch, run, return.

Permission Modes — Critical

!ADMIN! *dev* — Full access including br-ignore files. **Risk:** Can change core language (b-rite.md, error files). Use only for language updates, new *sf*/*sv*.

***dev* —** sv/sf file access, dev functions. **Risk:** May produce unwanted changes via interpretation, relaxed inference.

Regular (no prefix) — Strict syntax. AI halts on malformed scripts. Safe unless interMode or guessingMode are on.

interMode = 1 relaxes parsing; *guessingMode = 1* lets AI infer. Defaults: both 0.

Core b-rite Script Summary

Entry	Reads
`br`	functions + variables
`br{f}`	functions only
`br{v}`	variables only
`br{sf}`	system functions only
`*dev* !sdk|003| br`	SDK context (sdk-web)

`!sdk|003|` loads sdk-web from sdk-registry.br, reads brain-block, runs script in SDK context.

Top-Level Commands — New Project Setup

1. **First run:** Create Supabase project → run migrations (001, 002) → copy .env.example to .env.local with URL + anon key
2. `*dev* !sdk|003| br : { run; <npmInstall> } :` — Install deps
3. `*dev* !sdk|003| br : { run; <deployGuide> } :` — Supabase URL/key, .env.example

4. `*dev* !sdk|003| br : { run; <runFullPipeline> } :` — Full pipeline (spec → anchor → task → migration → build → audit → preview)
5. `*dev* !sdk|003| br : { run; <gitSetup> } :` — Push to GitHub (provide remote URL)

Or use `<masterProgramMaker>` for step-by-step agent-switch workflow.

SDK Functions (sdk-web)

Function	Purpose
<sdkWebSetup>	Setup instructions
<quickStart>	Project kickstart, folder structure
<specPhase>	A1 interview or A2 paste → ai/specs/
<anchorPhase>	Context anchors → ai/context/
<taskPhase>	Task breakdown → ai/tasks/
<migrationPhase>	Migrations → db/migrations/
<buildPhase>	Implement feature → app/, components/, lib/
<auditPhase>	Security, RLS verification
<previewPhase>	HTML previews → UI/
<runSinglePhase>	Run one phase by name
<runPhases>	Run selected phases
<runFullPipeline>	All phases in order
<masterProgramMaker>	Full workflow, agent-switch option
<agentHandoff>	Handoff template when declining agent
<deployGuide>	Git, Vercel, Supabase walkthrough
<createFeaturePack>	Scaffold feature pack
<runLocal>	npm run dev instructions
<npmInstall>	Run npm install in project root

<gitSetup>	Initial git + push (provide remote URL)
<projectVersion>	sdk-web and b-rite version info

SDK Variables

Variable	Path
aiSpecs	[root/ai/specs/]
aiContext	[root/ai/context/]
aiTasks	[root/ai/tasks/]
output	[root/UI/dev-output/]
dbMigrations	[root/db/migrations/]
setupProcedure	setup-procedure.txt
deployGuide, npmInstall, gitSetup	Function paths

Core b-rite Functions

Function	Purpose
sf <no-log>	Suppress logging
sf <log>	Detailed log entry
sf <delete>, <move>, <copy>, <new>	File ops (confirm delete)
sf <help>	Troubleshooting assistant
sf <install>	Add SDK/addon
sf <directory>	Search for feature/path
sf <index>	Check if exists
sf <archive>	Move to archived/
sf <fileIntegrity>	Scan protected files
sf <systemVersion>, <systemUpdates>	Version info, syntax updates

<start>	Onboarding, root, options, SDK install
<task>	Task creation flow
<commit>	Git stage, commit, push
<debug>	Cross-reference last error
<audit>	Full repo audit
<at>, <for>	Resolve location, apply to target
<lastPrompt>	Reference recent prompt
<nFunction>, <nVariable>	Create function/variable (devKit)
<nSDK>, <package>	Scaffold SDK, package (devKit)

Core Variables

Variable	Value
(root)	Project root
(tasks), (specs), (migrations)	Paths
(brain), (mLog), (options)	brain.md, changelog, options.br
(sdkRegistry)	sdk-registry.br (003 = sdk-web)

Human Interaction Guide

- **specPhase:** A1 = interview (AI asks); A2 = paste prompt (AI auto-fills)
- **deployGuide:** User provides Supabase URL and anon key
- **gitSetup:** User provides remote URL (create empty repo on GitHub first)
- **runFullPipeline / masterProgramMaker:** Optional B1 = reference paths for context
- **Delete/move:** AI always prompts confirmation

Without b-rite — Prompt from Docs Directly

If you prefer plain prompts over b-rite scripts:

> **Have AI read:** [root/ai/brain.md], [root/ai/sdks/sdk-web/sdk-web-brain-block.md], [root/ai/sdks/sdk-web/setup-procedure.txt], [root/ai/sdks/sdk-web/functions.br], [root/ai/sdks/sdk-web/variables.br]

Rule to follow: Paste sdk-web-brain-block.md content as context. It defines stack, paths, key functions. Use setup-procedure.txt for folder structure and phase order.

Removing the AI Component (Post-Dev)

app/, components/, lib/, db/ compile and run without ai/. The Next.js app is standalone.

To remove AI workflow:

1. Delete or ignore `ai/` folder
2. Keep app/, components/, lib/, db/, feature-packs/
3. README, .env.example, migrations, seed stay

You lose: b-rite scripts, spec-driven generation, AI phases. The deployed app is unchanged.

Generated from brain.md and sdk-web. Export: UI/dev-output/web-sdk-comprehensive-guide.html